Marco G. Barnfield

PhD Researcher, Machine Learning and Molecular Physics

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Summary

Third-year PhD researcher in the ExoMol group at University College London. I build machine learning tools for large molecular spectroscopy databases, used to model the atmospheres of exoplanets. I published two first-author papers in 2026, in the Journal of Molecular Spectroscopy and in JQSRT.

Experience

University College London, Department of Physics and Astronomy, Doctoral Researcher, Machine Learning and Astrophysical Spectroscopy — London, UK
Supervisors: Prof. Jonathan Tennyson FRS and Prof. Sergei N. Yurchenko — 2024 – present — 2 years

- Published a first-author paper on neural network corrections to the energy levels of rare isotopic molecules (Journal of Molecular Spectroscopy, 2026).
- Built a graph neural network pipeline that assigned quantum numbers to 224,650 unlabelled CO_2 states across 12 isotopologues (first author, JQSRT, 2026).
- Built an LLM-assisted pipeline (OCR, then markdown, then structured extraction) that turns published spectroscopy papers into machine-readable datasets, plus an agent that searches the literature for relevant papers.
- Built a PostgreSQL database and read-only REST API for MARVEL energy level data: 31 runs from 19 publications, covering 29 isotopologues.
- Lead graph neural network work on spectroscopic network solving (MARVEL_GNN) and on isotopologue energy level extrapolation.
- Rebuilt the spectroscopic network solving engine (MARVEL v5) with a Python backend and a React, TypeScript and D3.js frontend, using test-driven development.
- Maintain a 365-entry knowledge base of the ExoMol literature (ExoMol Wiki), with automated validation checks.

Fluorochem Ltd., Consultant and Scientific Content Writer — Remote — 2024 – 2025 — 1 year

- Gave consultancy on chemical synthesis and organometallics to support new product development and new website content.

University of Bristol, Royal Society of Chemistry Summer Research Student — Bristol, UK
Project: Machine Learning Emulator for Atmospheric Trace Gas Transport. Supervisor: Prof. Matt Rigby — June 2023 – Sept 2023 — 4 months

- Scaled a proof-of-concept ML emulator for greenhouse gas transport from its original domain to the whole UK and all regional methane measurement sites.
- Measured and improved the computational efficiency of the larger model on the BluePebble4 HPC cluster.
- Processed and plotted large spatial flux and emissions datasets with Python's xarray.

UCL Geography, London NERC DTP Summer Intern — London, UK
Project: Influence of mega constellations on our atmosphere. Supervisor: Dr Eloise Marais — June 2022 – Sept 2022 — 4 months

- Studied the NO_x and Al_2O_3 emissions from launching and maintaining satellite mega constellations.
- Gathered and analysed 3 large datasets from several sources in Python.
- Presented the work at the Natural History Museum, on the environmental impact of the new space race.

National Physical Laboratory, Junior Science Apprentice — Teddington, UK, 2017 – 2019, 2 years

- NiCE-MSi: optimised MALDI-MSi matrix preparation, and used DESI-MSi, the HTX TM-Sprayer and MATLAB. Helped the lab manager with daily operations and inventory.
- Nuclear Metrology: alpha and gamma spectroscopy, ICP-MS, and separation of ^{93}Zr and ^{93}Nb with anion exchange resin. Sorted and organised the safe storage and removal of nuclear waste.

Education

PhD **University College London**, Physics and Machine Learning — London, UK, 2024 – present

- Project: Machine Learning Isotopologue Extrapolation for use in Astronomical Spectroscopy
- Supervisors: Prof. Jonathan Tennyson FRS and Prof. Sergei N. Yurchenko
- Funded by the EPSRC Centre for Doctoral Training in Data Intensive Science and Industry (DISI)
- Postgraduate courses from Physics of Exoplanets to Statistical Data Analysis
- Chaired an external consultancy project building a machine learning pipeline to analyse flood defences in the USA (Fathom, ML research group industry placement)

MSci **University of Bristol**, Chemistry with Scientific Computing — Bristol, UK, 2019 – 2024

- First Class Honours
- Master's project: Photochemistry of Hot Water for Hot Jupiter Exoplanet Atmospheric Modelling. Supervisor: Prof. Andrew Orr-Ewing FRS
- First class average in years 1 and 2. Highest mark in the year for Advanced Data Science in year 3.
- Chaired the year 3 Group Project in Scientific Computing, which contributed to published software for photolysis rates of volatile organic compounds (In Silico Photochemistry Group, supervisor: Prof. Basile Curchod)
- Computing in C++ and Python: data analysis (pandas, NumPy), plotting (Matplotlib, Seaborn, Plotly), parallel computing (OpenMP, MPI, Cython, Numba), machine learning (scikit-learn) and SQL

National Physical Laboratory, Applied Science L3 BTEC Diploma (D*D*) and Laboratory Technician Standard (Distinction) — Teddington, UK, 2017 – 2019

Hampton School, 10 GCSEs (A* to B) — Hampton, UK, 2011 – 2016

Publications

Automated AFGL quantum number assignment for CO_2 isotopologues using a graph neural network — Dec 2026

First author. Graph neural network pipeline that labelled 224,650 previously unlabelled CO_2 states across 12 isotopologues.
M.G. Barnfield, S.N. Yurchenko, J. Tennyson
10.1016/j.jqsrt.2026.110111 (Journal of Quantitative Spectroscopy and Radiative Transfer 365, 110111)

Machine learning isotope shifts in molecular energy levels — July 2026

First author. Neural network corrections to isotopologue energy levels, improving more than 87% of CO_2 levels and, by transfer learning, more than 93% of CO levels.
M.G. Barnfield, O.L. Polyansky, S.N. Yurchenko, J. Tennyson
10.1016/j.jms.2026.112084 (Journal of Molecular Spectroscopy 418-419, 112084)

ExoMol line lists – LXIII. ExoMol line lists for 12 isotopologues of CO_2 Dec 2025
Co-author. Contributed the machine-learning AFGL quantum number assignments for the line lists.
S.N. Yurchenko, **M.G. Barnfield**, C.A. Bowesman, R.P. Brady, E.R. Guest, K. Kefala, Q-H. Ni, A.N. Perri, O.A. Smola, A. Solokov, C. Tao, J. Tennyson
10.1093/mnras/staf2135 (Monthly Notices of the Royal Astronomical Society 545(3), staf2135)

Leveraging Machine Learning for the Ariel Space Mission: A White Paper from the Ariel Machine Learning Working Group In review
Co-author.
K.H. Yip, **M.G. Barnfield**, et al.
(RAS Techniques and Instruments)

Talks

Contributed talk: "Machine Learning for Isotopologue Extrapolation in the ExoMol Project", Exoplanetary Atmospheres mini-symposium, High Resolution Molecular Spectroscopy (HRMS) 2026, Prague

Projects

Football transfer fee model 2026
End-to-end machine learning pipeline that estimates what a football transfer should cost, and where clubs over- or underpay. Python, SQL, XGBoost, Optuna, SHAP.

World Cup 2026 Elo model 2026
Elo ratings from 150 years of international results, used to measure which teams over- or under-performed in the group stage. Python, pandas, soccerdata.

English league pyramid Elo model 2026
Elo ratings for all four English professional tiers since 1993-94 (67,245 matches), benchmarked against bookmaker odds. Python, pandas, Plotly.

World Cup player embedding model (in progress) 2026
Web-scraping pipeline (397 players, 38 leagues) feeding a GraphSAGE player embedding model for scouting and valuation. Python, PyTorch Geometric, BeautifulSoup.

Technical Skills

Machine learning: Graph neural networks (GraphSAGE, PyTorch Geometric), PyTorch, uncertainty quantification, Hungarian-algorithm assignment, XGBoost with Optuna, SHAP, self-training with confidence gates, scikit-learn

Programming and data: Python (NumPy, pandas, SciPy, Matplotlib, Seaborn, Plotly, xarray), SQL, PostgreSQL, DuckDB, Git

Web and tooling: FastAPI, React, TypeScript, D3.js, Docker, test-driven development (pytest), config-driven experiments

HPC: SLURM clusters (UCL Hypatia, Bristol BluePebble4), Apptainer containers, OpenMP, MPI, Cython, Numba

LLM and agentic systems: Claude Code agentic workflows, LLM-based literature extraction pipelines, locally served open-weight models (vLLM), automated validation pipelines

Domain knowledge: Molecular spectroscopy (rotational and vibrational quantum numbers, isotopologues, spectroscopic networks), atmospheric chemistry, exoplanet atmospheres, the ExoMol database

Training

- Microsoft 365, STL Business Training
- MATLAB Onramp and Fundamentals, MathWorks
- Mental Health First Aid, MHFA England
- Radiation Protection Supervisor, RADMAN Associates
- Project Griffin, National Counter Terrorism Security Office

Other

- President and 1st Team Captain, University of Bristol Chemistry Football (2022 to 2024)
- Grade 8 Trumpet, Trinity Guildhall (2011)
- Custom PC building, diagnostics and repairs

References

Prof. Jonathan Tennyson FRS
Primary PhD supervisor. Massey Professor of Physics, Department of Physics and Astronomy, UCL. j.tennyson@ucl.ac.uk

Prof. Andrew Orr-Ewing FRS
MSci project supervisor. Professor, School of Chemistry, University of Bristol. A.Orr-Ewing@bristol.ac.uk